Google Summer of Code 2026 Proposal

Project Title:

AI-Powered Behavioral Analysis for Suicide Prevention, Substance Use, and Mental Health Crisis Detection with Longitudinal Geospatial Crisis Trend Analysis [AI4MH (AI for Mental Health Crisis Monitoring)]

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Synopsis

This proposal presents a structured framework for detecting and evaluating crisis signals from social media data in public health contexts. Social media posts are analyzed at the county level using sentiment intensity, volume spikes, geographic clustering, and discussion patterns. It improves reliability by accounting for bot activity, media-driven spikes, and data sparsity, and produces a confidence-adjusted crisis score. Based on the predefined thresholds, signals are prioritized and routed to human reviewers for final decision-making.

Crisis Signal Design

The Crisis Score is calculated from the social media posts within a 72 Hour Window for each county. Since the data extraction from the social media and lexicon filtering of posts has been completed in the prototype phase, I want to also include the extraction of news headlines and news articles from Google News and Google Trends to strengthen the crisis signal design.

Before moving to calculate the crisis score, the posts must satisfy minimum sample size barrier.

1. Minimum Sample Size Barrier

N_C — Number of posts from a county within the window
N_B — Minimum sample size Barrier

Case 1: $N_C \geq N_B$

In this case, the posts from the county will be allowed to calculate the crisis score.

Case 2: $N_C < N_B$

In this case, instead of depending only on the current window, the previous 72-hour window of the county will also be included, and if the combined window crosses the minimum sample size barrier, then, it would be analyzed with low confidence flagged.

Crisis Score Calculation

The raw crisis score (C) for a given county over a 72-hour window will be a composite of three factors: Sentiment Intensity (I), Volume spike (V), Geographical clustering (K).

$$\mathbf{C = 0.5\,I + 0.3\,V + 0.2\,K}$$

Where:

- **Sentiment Intensity (I):**
  It measures the severity of crisis-related language within the 72-hour window.

$$I = \frac{\sum_{p=0}^{n}(s_p + r_p)}{n}$$

  Where:
  - s_p is a normalized sentiment score of a post.
  - r_p is a normalized risk factor of the post. (e.g. high risk-1, moderate risk post-0.60, low risk-0.20)

Sentiment Intensity is smoothed using an exponential moving average method. It is normalized using the standard **Min-Max Normalization** method.

- **Volume Spike (V):**
  It measures deviation of high-risk post count from historical baseline

$$Z = \frac{N_{current} - \mu_{baseline}}{\sigma_{baseline}}$$

- $N_{current}$ - number of high-risk posts detected in 72-hour window for a specific county.
- $\mu_{baseline}$ - average number of high-risk posts in each window over previous 10 windows (i.e. 30 days).
- $\sigma_{baseline}$ – standard deviation measures how much high-risk post volume fluctuates between windows.
- Volume spike score is normalized using **Min-Max Normalization** method.

- **Geographic Clustering (K):**

$$K = \frac{A_c}{T_c}$$

Where:

- A_c is the number of active adjacent counties.
- T_c is the total number of adjacent counties.
- The county is active if:
- $$Z > 2 \; and \; I > 0.6$$
- The adjacent counties will be determined using coordinates of the county.

Weights Distribution

- The Sentiment Intensity is given higher weight because content and language are the primary indicators of distress.
- Volume Spike is weighted second highest since it confirms that distress is being expressed by multiple community members, not just by an individual.
- The Geographical Clustering is weighted lowest because while regional spread adds confidence, but a genuine crisis can occur within a single county. Edge case handled where k=0, a severe isolated crisis can still achieve C=0.8, which meets high risk level thresholds and ensures human review.

Bias and Mitigation

1. **Bot Amplification or Coordinated Activity**

Initially, I thought bot detection may appear to be a classification problem which can be handled using an advanced machine learning model. However, it fails to capture coordinated behavior. In practice, bot activity must be analyzed collectively rather than individually. Hence, bot amplification is now addressed at the cluster level.

The proposed bot amplification detection system works as follows:

- Posts are clustered using a simple Density-based clustering model based on the content similarity.
- Then top K clusters are selected based on size.
- For each cluster, a rule-based bot score is computed based on the following heuristics:
  - **Account age:** Newly created accounts are assigned to higher risk scores.

- **Verification status:** Unverified accounts contribute positively to the bot score.
- **Reddit employee status**: Verified internal accounts are treated as more reliable, while others contribute slightly to bot scores.
- **Temporal behavior**: Accounts posting a lot of posts in a short time interval are more likely to be flagged as bots.
- **Content Similarity**: Accounts posting a lot of identical posts are more likely to be flagged as bots.

Individual Bot score = (Account Age Score * 0.10) +

(Verification Status Score * 0.15) +

(Temporal Behavior Score * 0.30) +

(Content Similarity Score * 0.45)

Where:

- Account Age Score: 1 if <30 days, 0.5 if 30-90 days, 0 if >90 days
- Verification status: 1 if unverified, 0 if verified
- Temporal Behavior: Posting frequency / expected human frequency
- Content Similarity: % of posts identical to each other

Temporal Behavior is given more weight because posting at a high rate is more likely to be bots and next priority is given to the content similarity of the account posts followed by verification status.

Cluster bot score = Average bot score of all users in the cluster.

- Final bot amplification score (B) is the average cluster bot score of all selected clusters and if it is greater than the threshold, then bot amplification is flagged.

This approach enables the system to detect coordinated activity much more effectively without relying on an advanced machine algorithm.

2. Media-driven spikes

Media-driven spikes occur when external events or news coverage leads to a sudden increase in related posts. These spikes may not necessarily represent organic distress but rather widespread attention due to media influence.
Since the media-driven spike is also concentrated around specific topics, the same content-based clusters are used which have been generated for bot detection.

The system detects media-driven spikes using the following approach:

- Posts are clustered using a simple Density-based clustering model based on the content similarity.
- Then top K clusters are selected based on size.
- For each cluster, a semantic similarity score is calculated between the cluster post content and recent news headlines or trending topics (e.g., Google Trends, Google News).

Cluster media score = Average similarity scores of all posts in the cluster.

- Final media-driven score (M) is the average Cluster media scores of all selected clusters and if the score exceeds the threshold, then media-driven spike is detected.

3. Rural Underrepresentation or Sparse Data

In some counties, especially rural regions, the available posts may be sparse. This can lead to underrepresentation of actual conditions and unreliable crisis signals.

To address this issue, a population factor is introduced;

$$P_f = \sqrt{\frac{P_{avg}}{P_c}}$$

- P_c = population of the county
- P_{avg} = average population across all counties

Uncertainty due to sparse data:

$$U = \frac{1}{1+P_f}$$

- Smaller population → higher uncertainty

Confidence Score

This score integrates multiple sources of uncertainty, including bot amplification, media-driven spike, and sparse data conditions.

The confidence score is defined as:

$$Confidence = 1 - (\,0.4\,B + \;0.3\,M + \;0.3\,U\,)$$

Where:

- B – Final bot amplification score
- M - Final media-driven score
- U – Uncertainty due to sparse data

- Bot activity (0.4) weighted highest because coordinated manipulation poses greatest risk
- Media spikes (0.3) and sparse data (0.3) weighted equally

Final Score Adjustment

Final Crisis Score (C_f) = Raw Crisis Score (C) × Confidence

This final crisis score ensures that high-intensity signals are only considered when supported by sufficient confidence and reducing the likelihood of false positives in public health monitoring.

Escalation Thresholds

The escalation thresholds for the system depends on the Final Crisis Score and Confidence

The following thresholds are defined:

- **High-risk Level:**
  If $\boldsymbol{C_f > 0.7}$ **and Confidence > 0.7**, the signal is classified as a high-priority case and provided for human review with high priority recommendation.
- **Moderate-risk Level:**
  If $\boldsymbol{C_f > 0.5}$, the signal is classified as a moderate risk case and provided for a human review.
- **Low-risk Level:**
  If $\boldsymbol{C_f \leq 0.5}$, the signal is considered low-risk and is monitored for upcoming windows.

Human-in-the-loop Process

The system is designed such that all escalation actions require human validation to ensure accountability and reduce the risk of false positives.

When a signal falls into either the high-risk or moderate-risk category, it is routed to human reviewers.

The high-risk level is provided with the high priority recommendation.

Once the county is classified either of the two cases the reviewers are provided with:

- A high-level summary containing the final crisis score, confidence score, uncertainty due to sparse data, risk level, geographic location, sample size and time window.
- Past trends of the county such as crisis score variation, sentiment intensity trend, and post volume trend.
- Bias Detected:
  If Bias Detected the following information are provided for each bias,

Bot Amplification Flagged

- Cluster-level insights (such as top K cluster sizes, number of clusters formed, sample posts from each cluster).
- WordCloud of top k clusters to provide contextual understanding of the signal.
- Bar Plots of average posting frequency of each cluster and cluster bot score.

Media-driven spike detected

- Cluster-level insights (such as top K cluster sizes, number of clusters formed, sample posts from each cluster).
- WordCloud of top k clusters to provide contextual understanding of the signal along with recent google Trends and News headlines.
- Bar Plots of cluster media-driven score.

Audit Logging Structure

To ensure transparency, the system maintains a comprehensive audit log of all scores and details.

1. Timestamp
2. County
3. Sample size
4. Calculated Scores: Sentiment Intensity (I), Volume Spike (V), Geographic Clustering (K)
5. Raw Crisis Score (C)
6. Confidence Score
7. Final Crisis Score
8. Risk Level
9. Bias Identified
    - Final bot score
    - Final media-driven score
    - Uncertainty due to sparse data
10. Human Review
    - Reviewer Id
    - Decision
    - Notes

Governance Reflection

Primary Risk of Premature Deployment

The primary risk of premature deployment is false crisis signals caused by amplification bias. Coordinated bot activity, media-driven spikes, or sudden increase in online discussions may increase the severity of crisis. If such signals are treated as crisis without sufficient validation, they may lead to unnecessary escalation, misallocation of public health resources, and affect the trust in the system.

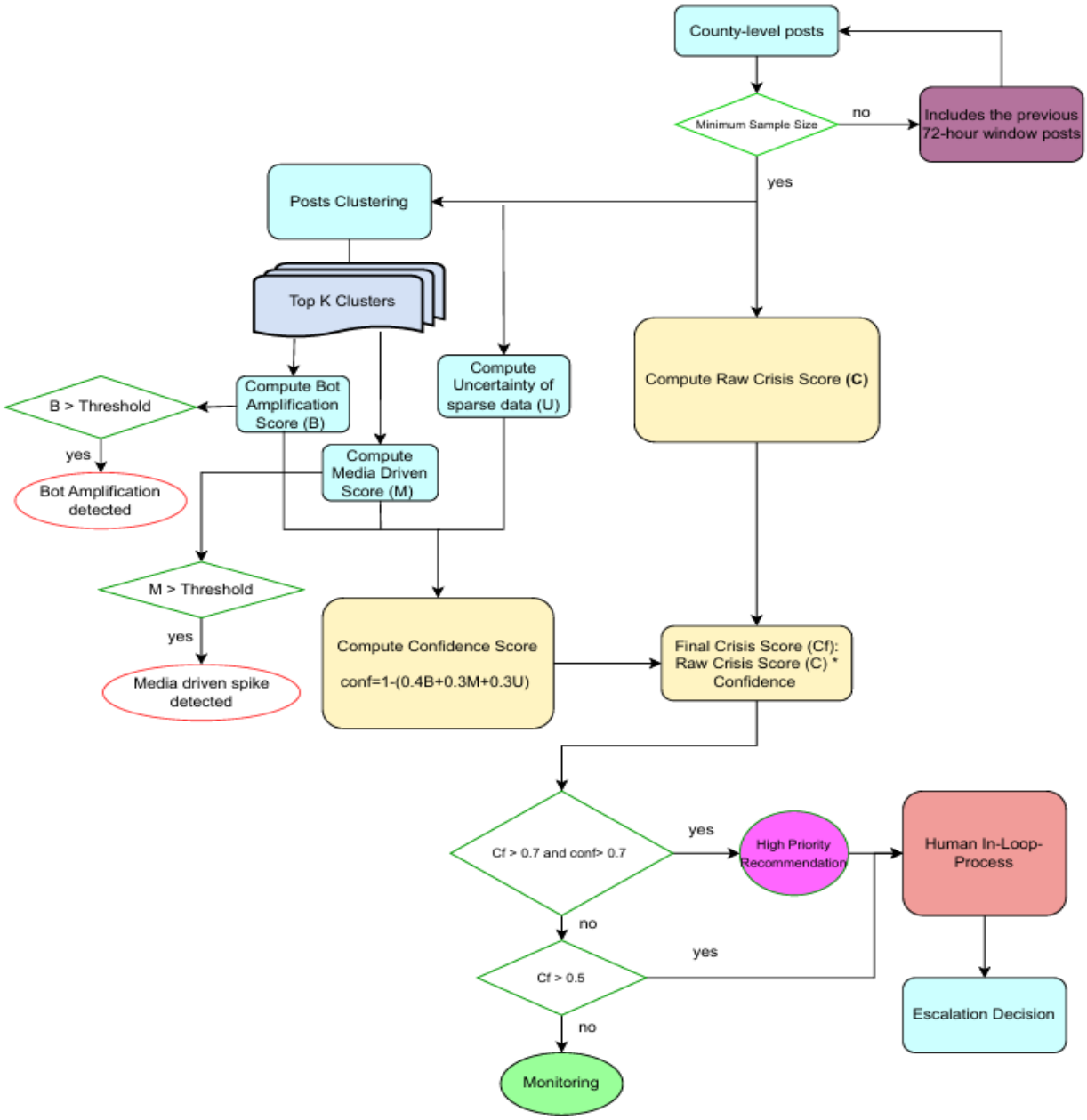
Most Important Safeguard

The most critical safeguard is the incorporation of a human-in-the-loop validation process. All the risk signals are provided to human reviewers along with the evidence for signals such as details of all the scores, bias detection details and the evidence for that and also sample posts are provided to ensure that final decisions are made with contextual understanding and expert judgement, thereby reducing the risk of acting on manipulated signals.

Pseudocode

```
FUNCTION crisis_signal_pipeline(posts, county):

    # Step 1: Sample Size Barrier

    IF size(posts) < MIN_SAMPLE_SIZE:
      # Extend window
       extended = merge_previous_window(posts)
       IF size(extended) < MIN_SAMPLE_SIZE:
           Return "INSUFFICIENT_DATA"
       ELSE:
        posts=extended
        sparse_penalty=0.2 # Reducing the confidence due to sparse data
    ELSE:
       Sparse_penalty=0  # Minimum Sample Size Barrier Satisfied

 # Step 2: Compute Raw Crisis Score

    sentiment_score = calculate_sentiment(posts)
    volume_spike = calculate_volume_spike(posts, baseline_30days)
    geographic_cluster = count_active_adjacent_counties(county)

    crisis_score = (0.5 × sentiment_score) +
                   (0.3 × volume_spike) +
                   (0.2 × geographic_cluster)

 # Step 3: Detect biases

    clusters = group_similar _posts(posts)
    top_clusters = select_top_k(clusters, K)
    B = compute_bot_score(top_clusters)
    M = compute_media_score(top_clusters)
    U = compute_uncertainty(N, county_population, sparse_penalty)

    if B > BOT_THRESHOLD:
        bot_flag = True
    else:
        bot_flag = False

    if M > MEDIA_THRESHOLD:
        media_flag = True
    else:
        media_flag = False

    # Step 4: Apply confidence adjustment
    confidence = 1 - (0.4 * B + 0.3 * M + 0.3 * U)
    C_final = C * confidence   # final score

    route_to_human = Escalation_Thresholds(C_final, confidence)

    if route_to_human:
        human_decision = human_review_interfac()
        final_action = human_decision

    else:
        final_action = "MONITOR"

    log_event()
    return final_action
```

Flowchart for Crisis Signal Design



About Me

I am an Engineering Student pursuing Mining Engineering from Indian Institute of Technology (BHU) Varanasi. I am deeply interested in applying AI systems to solve real-world problems and have developed multiple projects addressing practical use cases.

One of my projects, **MedSnap**, is an AI-assisted medical report summarization tool designed for medical students and researchers. It simplifies complex medical documents into concise, understandable notes and provides real-time updates on relevant medical news, enabling efficient learning and research.

I have also developed an **Indigenous AI Chatbot** for Indian mining workers and mining exam aspirants. This system is trained on Indian mining rules and regulations, including official acts and safety guidelines. It assists users by providing accurate safety measures and actionable guidance for various mining scenarios, contributing to improved awareness and safety practices.

I am highly motivated to work on this project and have already developed a working prototype of the dashboard. I am now eager to extend this work to social media post analysis, ensuring that the system is ethical, reliable, and aligned with real-world requirements.